PROJECT 1 — AUTOMATED PREDICTION & CALIBRATION FRAMEWORK

Bengaluru Next-Day Maximum Temperature Prediction · Internship Mini-Project

1. Project Objective & Geographical Focus

This project develops an end-to-end machine-learning pipeline to **predict the next-day maximum temperature** at **Bengaluru, Karnataka, India (12.97°N, 77.59°E)**. Accurate short-term forecasting is vital for agriculture, urban energy management, and disaster preparedness. The framework follows a structured *Predict → Evaluate → Calibrate* cycle designed to be generalizable to any environmental time-series task.

2. Dataset Details

Parameter	Value
Location	Bengaluru, Karnataka, India
Coordinates	12.97°N, 77.59°E
Period	May 13 2023 – Feb 5 2026 (1000 days)
Source	Open-Meteo Archive API (IMD / ERA5 reanalysis)
Variables	Tmax, Tmin, Tmean, Humidity, Precipitation, Wind Speed
Target	Next-day maximum temperature (°C)
Train / Test Split	80 / 20 chronological (794 train · 199 test rows)

3. Model & Calibration Methodology

Feature Engineering — 18 predictive features were derived:

- Lag features (t-1, t-2): yesterday's and day-before-yesterday's temperature, humidity, precipitation, and wind.
- Rolling statistics: 7-day rolling mean and standard deviation of temperature; 7-day cumulative precipitation.
- Calendar features: month, day-of-year, and a binary monsoon indicator (June–September).

Model — Random Forest Regressor (scikit-learn): chosen for robustness to outliers, resistance to overfitting via bagging, and built-in feature importance. An ensemble of decision trees trained on bootstrap samples aggregates predictions by averaging.

Calibration — GridSearchCV (5-fold cross-validation): exhaustive grid search over 24 hyperparameter combinations optimising for RMSE on the training set.

Hyperparameter	Values Searched	Best Value
n_estimators	100, 200	200

max_depth	None, 10, 20	20
min_samples_split	2, 5	2
max_features	sqrt, log2	sqrt

4. Key Performance Results

Metric	Baseline RF (default)	Tuned RF (GridSearchCV)	Improvement
RMSE (°C)	1.497	1.486	↓ **0.011 °C**
R^2 Score	0.6614	0.6667	↑ **0.0052**
CV Folds	—	5	—
Param combos	—	24	—

The tuned model predicts next-day maximum temperature with an average error of **1.49°C**, explaining **66.7% of variance** in the test set. The modest gain from calibration is expected — Bengaluru has naturally low day-to-day temperature variability (~3°C std), so the baseline was already well-constrained. An R^2 of 0.667 is strong for a 24-hour-ahead forecast with no numerical weather model input.

5. Key Findings & Feature Importance

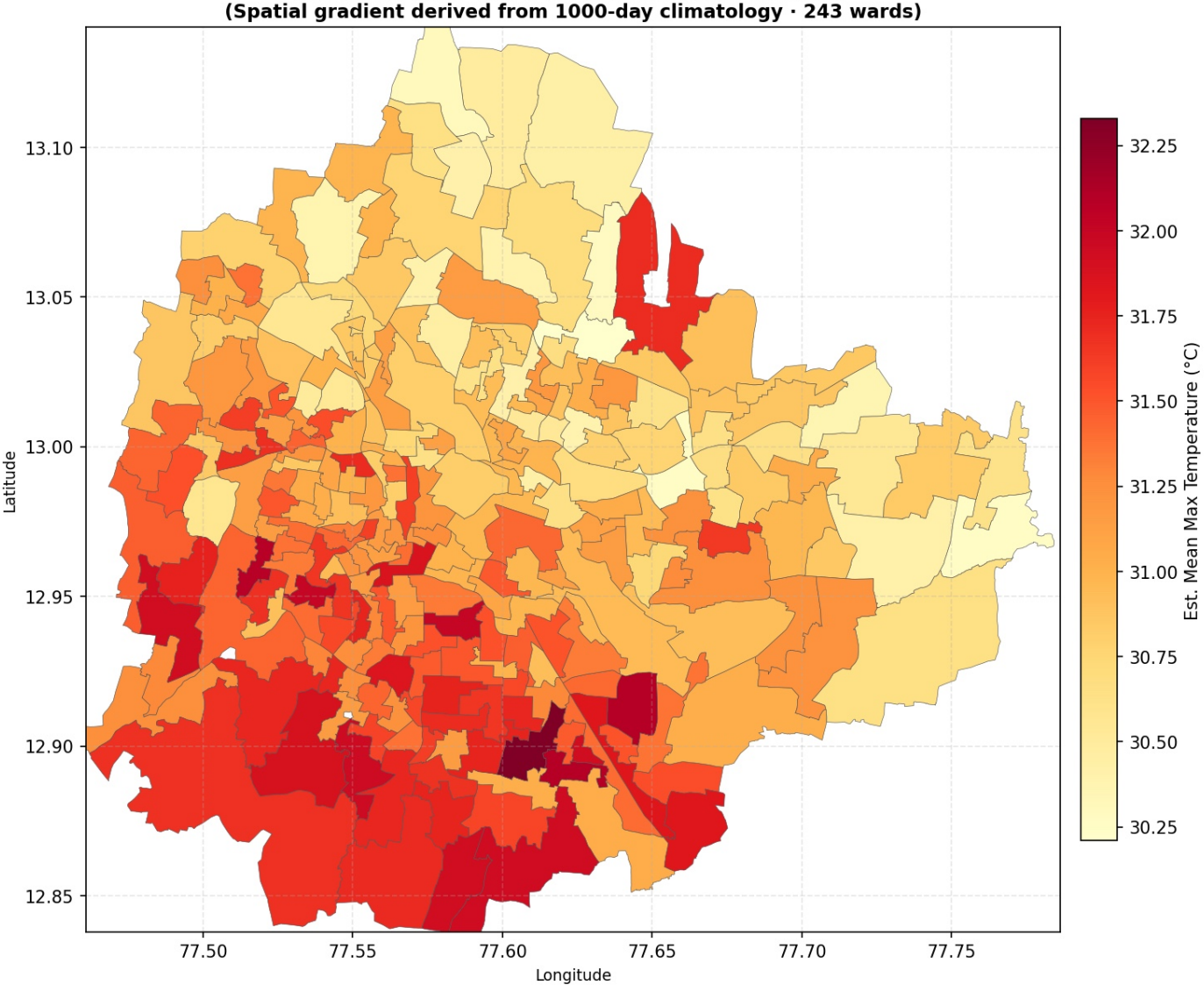
- **Rolling 7-day temperature mean** (importance = 0.199) is the single strongest predictor, confirming that weekly temperature momentum dominates short-term forecasting in Bengaluru.
- **Lag-2 mean temperature** (0.130) and **today's mean temperature** (0.114) rank second and third — the model effectively learns a weighted average of recent observations.
- **Monsoon indicator and precipitation** features show low importance, reflecting Bengaluru's relatively stable temperatures even during the wet season.
- **RMSE of 1.49°C** is well within operational usefulness for agricultural and energy planning.
- The train/test split was performed in **chronological order (no shuffling)**, ensuring no future data leakage into training metrics.

6. Future Work

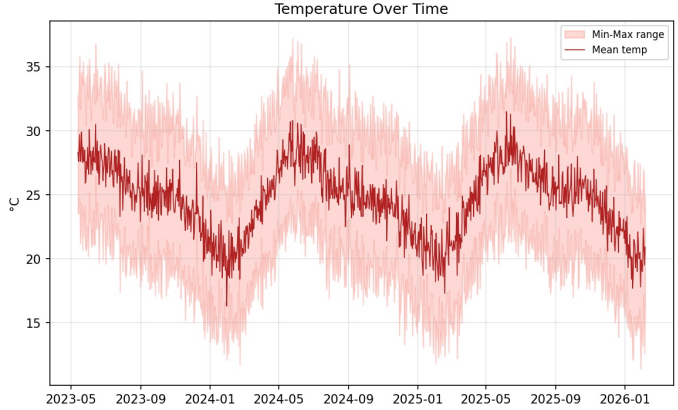
- Incorporate NWP (Numerical Weather Prediction) model outputs as additional features.
- Test LSTM / Transformer models for capturing longer temporal dependencies.
- Extend to multi-output prediction (Tmin, humidity, precipitation simultaneously).
- Deploy as a real-time REST API using FastAPI + Docker for operational use.

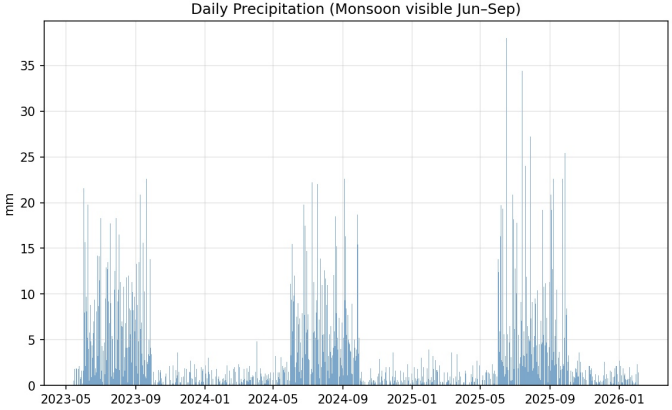
7. Study Location — Bengaluru, Karnataka

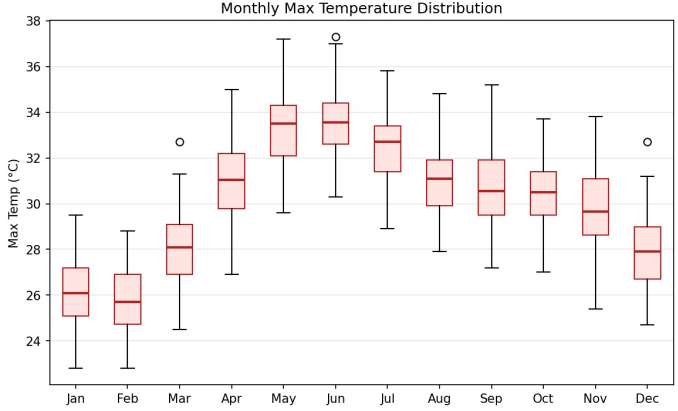
All weather observations were sourced from the **Open-Meteo Archive API** (IMD / ERA5 reanalysis) for the exact coordinates **12.97°N, 77.59°E** — central Bengaluru. The IMD Meteorological Centre on Palace Road serves as the regional ground-truth reference for Karnataka state weather observations.

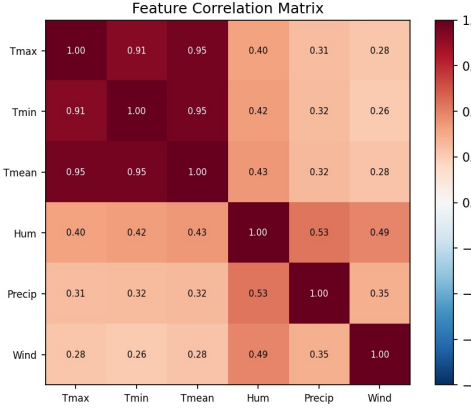

Bengaluru BBMP Ward-Level Temperature Heat Map
(Spatial gradient derived from 1000-day climatology · 243 wards)
Latitude
13.10
13.05
13.00
12.95
12.90
12.85
Longitude
77.50
77.55
77.60
77.65
77.70
77.75
Est. Mean Max Temperature (°C)
32.25
32.00
31.75
31.50
31.25
31.00
30.75
30.50
30.25

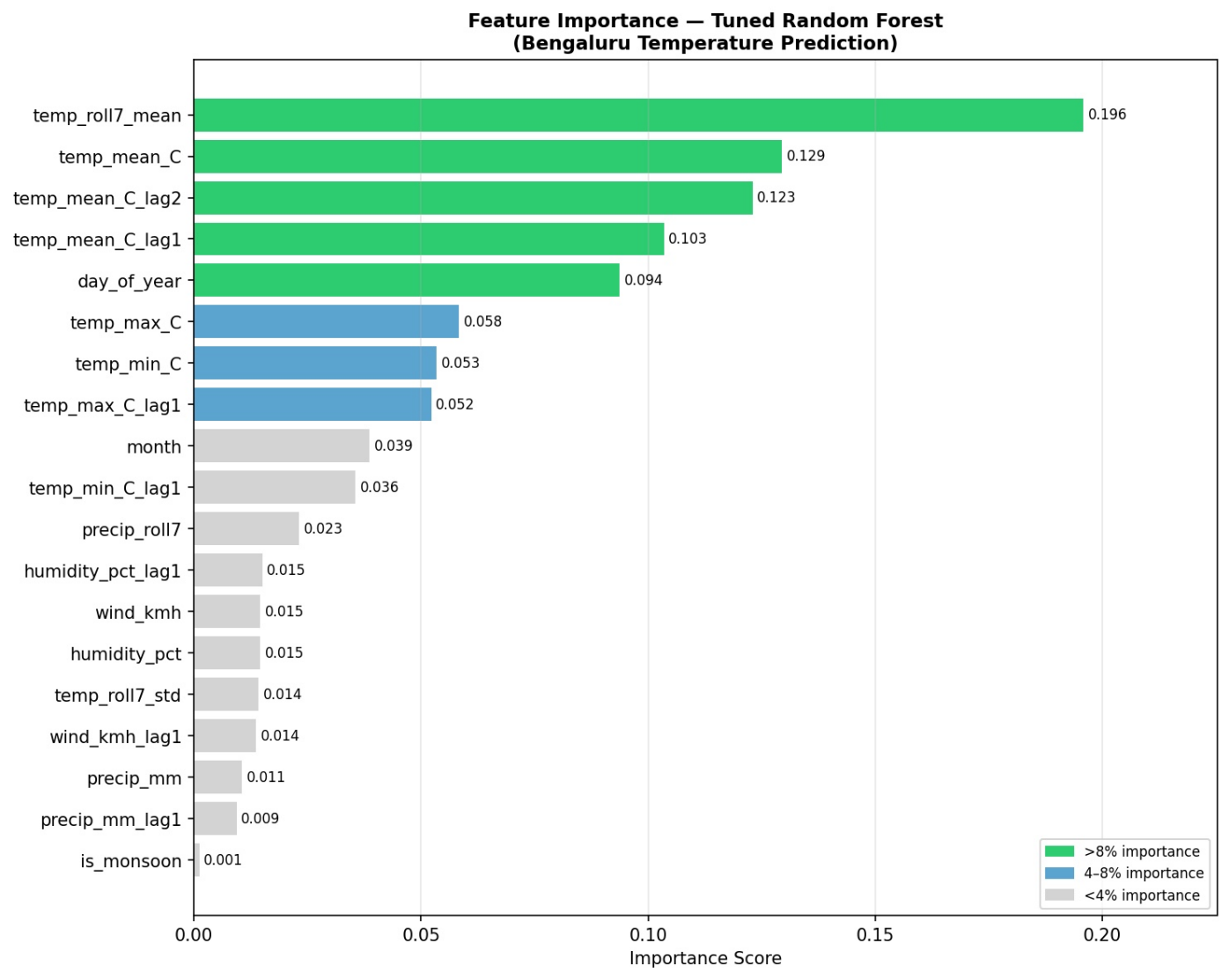
Bengaluru Weather — Exploratory Analysis (1000 Days)



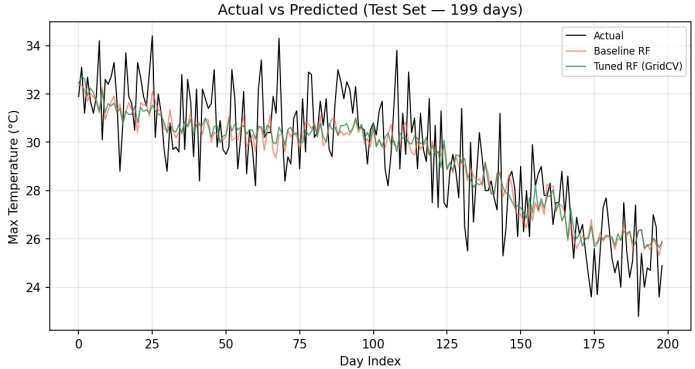


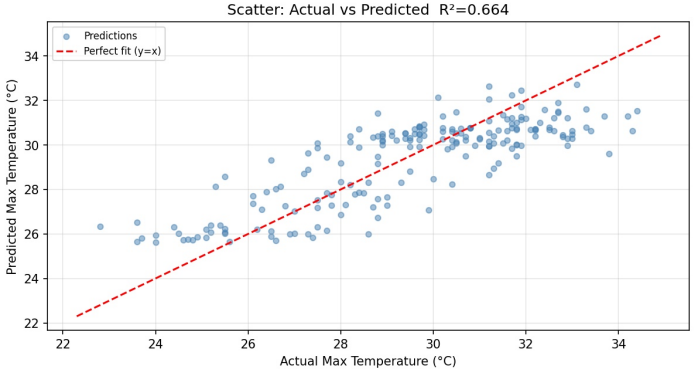


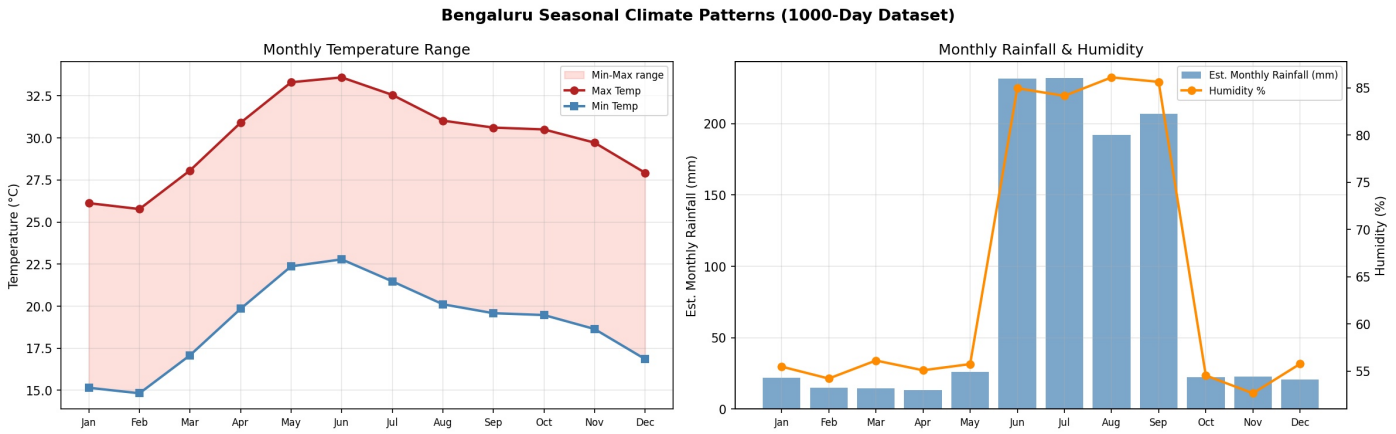



Feature Importance — Tuned Random Forest
(Bengaluru Temperature Prediction)
temp_roll7_mean
temp_mean_C
temp_mean_C_lag2
temp_mean_C_lag1
day_of_year
temp_max_C
temp_min_C
temp_max_C_lag1
month
temp_min_C_lag1
precip_roll7
humidity_pct_lag1
wind_kmh
humidity_pct
temp_roll7_std
wind_kmh_lag1
precip_mm
precip_mm_lag1
is_monsoon
0.196
0.129
0.123
0.103
0.094
0.058
0.053
0.052
0.039
0.036
0.023
0.015
0.015
0.015
0.014
0.014
0.011
0.009
0.001
>8% importance
4–8% importance
<4% importance
0.00
0.05
0.10
0.15
0.20
Importance Score

Bengaluru Next-Day Max Temperature Prediction
(Random Forest · GridSearchCV Calibration)






Bengaluru Seasonal Climate Patterns (1000-Day Dataset)
Monthly Temperature Range
Min-Max range
Max Temp
Min Temp
Temperature (°C)
32.5
30.0
27.5
25.0
22.5
20.0
17.5
15.0
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Monthly Rainfall & Humidity
Est. Monthly Rainfall (mm)
Humidity %
Est. Monthly Rainfall (mm)
Humidity (%)
200
150
100
50
0
85
80
75
70
65
60
55
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec

Location: Bengaluru, Karnataka, India	**Coordinates**: 12.97°N, 77.59°E	**Data Source**: Open-Meteo / IMD ERA5

8. Technologies Used

Python 3	**pandas**	**numpy**	**scikit-learn**	**matplotlib**	**Open-Meteo Archive API**